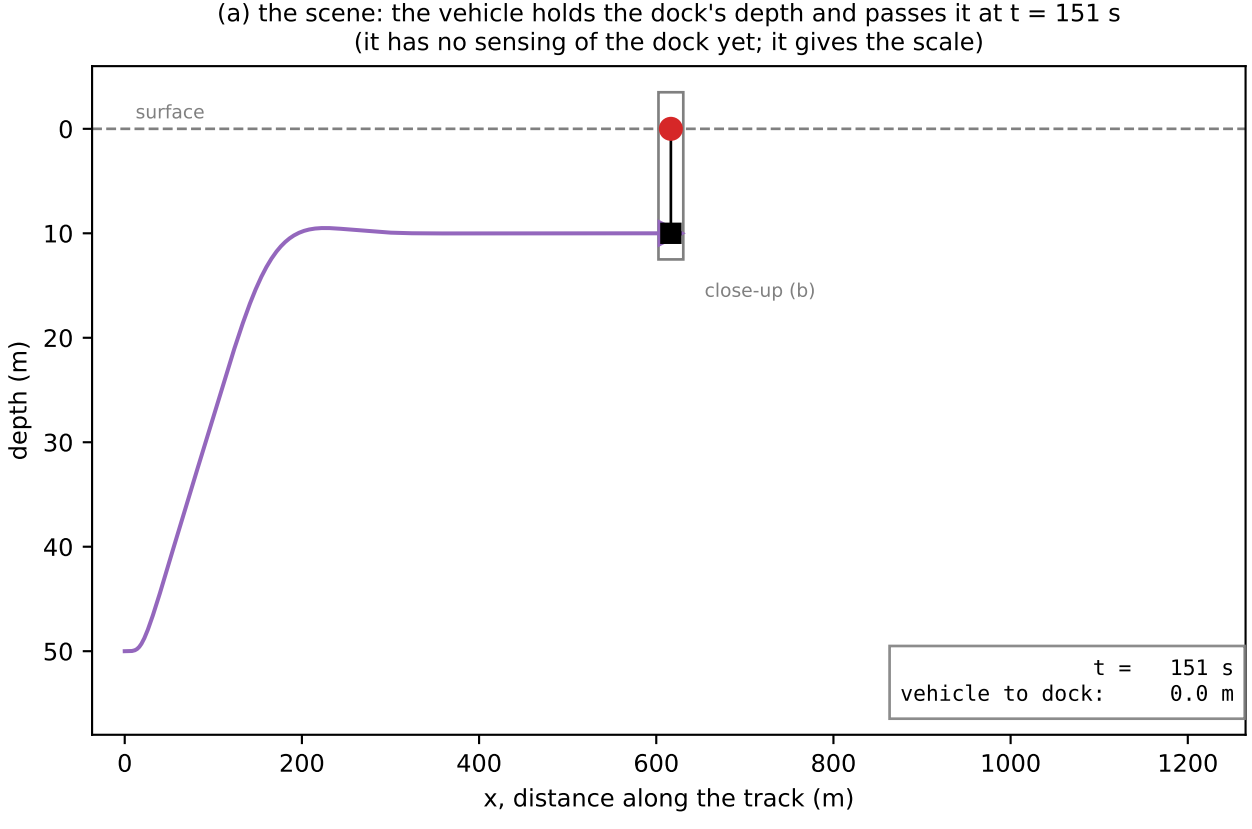
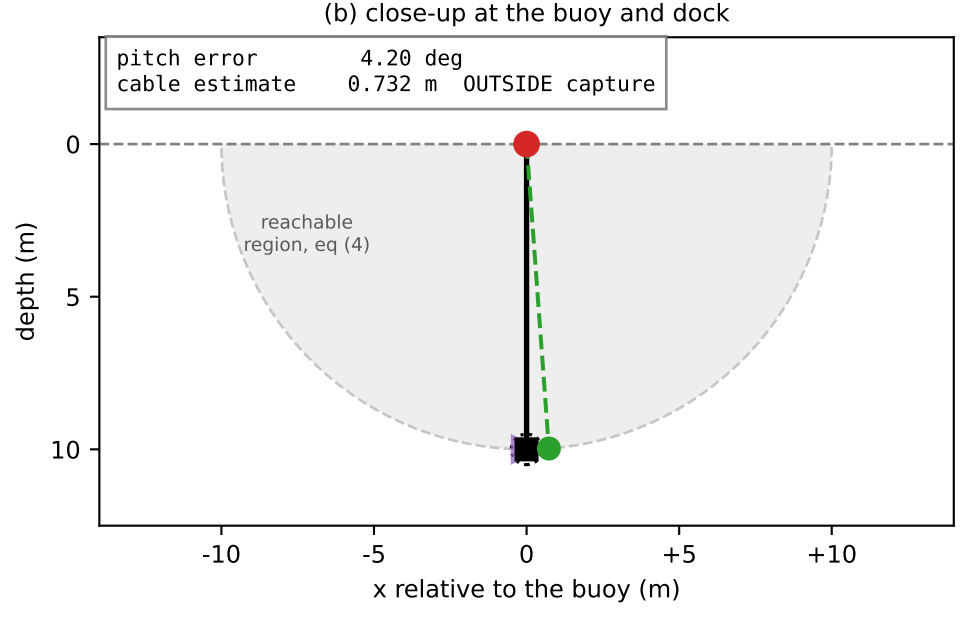
Where is the dock? Its navigation unit has a consumer MEMS, 100 deg/hr gyro bias.
Black is the truth. Green is the dock position estimated from that unit's attitude through the cable, eq (3).




buoy: source, position known
true dock and cable (never moves)
dock estimate from cable, eq (3): $p_S + L\,C_b^n u^b$
reachable region, eq (4): $\|p_D - p_S\| \le L$
capture radius 0.5 m
vehicle